DevBoard - Design Document

CS5610 Web Development

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1. Project Description

Overview

DevBoard is a community-driven project showcase and discovery platform built for developers and CS students. Every developer builds dozens of projects throughout their academic and professional journey, but there is no dedicated lightweight platform to showcase them, discover what peers are building, or get meaningful feedback. LinkedIn is too formal, GitHub is too technical, and portfolios are static.

DevBoard fills this gap by providing a living portfolio feed where developers can post their projects with tech stack tags, live demo links, and descriptions, while peers can bookmark projects they find interesting and leave structured feedback with ratings. It serves as a centralized hub for the developer community, part showcase, part discovery engine.

Key Features

- **Project Listings** - Create, edit, delete, and browse developer projects with title, description, tech stack tags, GitHub link, live demo URL, and category
- **Search & Filter** - Search projects by title or description, filter by tech stack and category for quick discovery
- **Community Feedback** - Leave structured feedback with a 1–5 star rating and comment on any project
- **Bookmarks** - Save interesting projects for later with optional personal notes
- **Community Stats** - View trending tech stacks and most active project categories across the platform
- **User Authentication** - Register and login with Passport.js to manage personal bookmarks and feedback
- **Pagination** - Browse large datasets efficiently with paginated views (10 per page) across projects, feedback, and bookmarks

Technology Stack

- **Frontend:** React, CSS (component-scoped), PropTypes
- **Backend:** Node.js, Express.js
- **Database:** MongoDB Atlas (native driver, no Mongoose)
- **Authentication:** Passport.js with Local Strategy, express-session
- **Tools:** ESLint, Prettier
- **Deployment:** Vercel (frontend), Render (backend)

Goals

- Give developers a dedicated space to showcase projects beyond static portfolios and GitHub repos
- Enable peer discovery, see what others in the community are building
- Provide meaningful, structured feedback with ratings to help authors improve
- Surface community trends through tech stack and category analytics
- Create a lightweight, fast platform that does one thing well

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2. User Personas

Persona 1: Alex - The CS Student

- **Age:** 21 | **Year:** Junior
- **Goals:**
  - Showcase projects for internship applications beyond a static portfolio site
  - Get peer feedback to improve project quality before interviews
  - Discover what other students are building for inspiration
- **Pain Points:**
  - Static portfolio sites feel outdated quickly and require manual maintenance
  - GitHub is too code-focused, non-technical recruiters can't evaluate project quality
  - No easy way to get structured feedback from peers on completed projects
- **How DevBoard Helps:**
  - Creates project listings with descriptions, tech stacks, and live demo links
  - Receives star ratings and comments that highlight strengths and areas to improve
  - Browses and searches other students' projects for inspiration and benchmarking

Persona 2: Jordan - The Hiring Manager

- **Age:** 34 | **Role:** Engineering Manager at a mid-size tech company
- **Goals:**
  - Discover what students and junior developers are actually building
  - Filter by tech stack to find candidates with relevant experience (e.g., React, Python)

    - See live demos and project descriptions rather than just GitHub repos
- **Pain Points:**
    - Resumes all look the same, everyone lists the same technologies
    - GitHub profiles require deep diving into code to evaluate quality
    - No easy way to browse student projects by technology or category
- **How DevBoard Helps:**
    - Filters projects by tech stack (React, Node, Python, etc.) and category (Web, AI, Mobile)
    - Bookmarks promising candidates' projects with personal notes for follow-up
    - Reviews community feedback and ratings to gauge project quality at a glance

Persona 3: Priya - The Curious Developer

- **Age:** 27 | **Role:** Mid-level Full Stack Developer
- **Goals:**
    - Explore interesting projects across different categories and tech stacks
    - Bookmark projects to revisit later for reference or inspiration
    - Leave constructive feedback to help fellow developers improve their work
- **Pain Points:**
    - Product Hunt is too startup/business focused for developer side projects
    - GitHub trending is only about stars and popularity, not project quality or feedback
    - No lightweight platform exists for casual project discovery with peer reviews
- **How DevBoard Helps:**
    - Browses projects by category (Web, AI, DevTools, Game) and searches by keyword
    - Bookmarks projects with personal notes like "Great reference for my next project"
    - Leaves detailed feedback with 1–5 star ratings to help authors improve

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3. User Stories

Preshit Ravindra Pimple - Project Listings & Tech Stack (Collection: projects)

Story 1: Create Project Listing As a developer, I want to create a project listing with title, description, tech stack tags, GitHub link, live demo URL, and category, so others can discover what I have built.

- Form validates all required fields (title, description, tech stack, category, author)
- Tech stack entered as comma-separated values and stored as an array
- Project appears in the list immediately after submission

Story 2: Edit or Delete Projects As a developer, I want to edit or delete my project listings, so I can keep my showcase accurate and up to date.

- Edit form pre-fills all existing data
- Delete removes the project immediately
- Changes reflect in the project list without page reload

Story 3: Browse and Filter Projects As a developer, I want to browse all projects and filter by tech stack, category, and date, so I can discover relevant projects quickly.

- Filter dropdowns for tech stack (React, Node, Python, etc.) and category (Web, Mobile, AI, etc.)
- Filters apply immediately and reset page to 1
- Results paginated at 10 projects per page

Story 4: Search Projects As a developer, I want to search projects by title or description, so I can find specific projects without scrolling through the entire list.

- Search input with real-time filtering
- Searches across both title and description fields
- Works in combination with tech stack and category filters

Story 5: View Project Detail As a developer, I want to view a project detail page with full description, tech stack, and links, so I can explore projects in depth.

- Shows all fields: title, author, category, description, tech stack tags, GitHub link, demo URL, date
- GitHub and demo links open in new tab
- Back button returns to project list

Story 6: View Community Stats As a developer, I want to see stats such as most used tech stacks and most active categories, so I can understand trends in the community.

- Top 5 tech stacks displayed with project counts
- All categories displayed with project counts
- Stats sorted by count descending

Gurudatt Pramod Gaonkar - Bookmarks & Community Feedback (Collections: bookmarks, feedback)

Story 1: Bookmark Projects As a developer, I want to bookmark projects I find interesting with an optional note, so I can revisit them later.

- One-click bookmark from project detail page
- Optional note field for personal context (e.g., "Good reference for my portfolio")
- Bookmark saved with logged-in username automatically

Story 2: Manage Bookmarks As a developer, I want to view all my bookmarked projects in one place with edit and delete options, so I can manage my saved content.

- Bookmark list shows project title (resolved from ID), personal note, and date
- Edit note inline with save/cancel buttons
- Delete removes bookmark immediately
- Paginated at 10 bookmarks per page

Story 3: Leave Structured Feedback As a developer, I want to leave structured feedback on a project with a rating (1–5 stars) and comment, so the author gets meaningful input.

- Rating select with 1 (Poor) through 5 (Excellent) options
- Comment textarea for detailed feedback
- Feedback appears immediately under the project detail page

Story 4: Edit or Delete Feedback As a developer, I want to edit or delete my feedback, so I can update my opinion if my view changes.

- Edit button opens form pre-filled with existing rating and comment
- Delete removes feedback immediately
- Changes reflect in the feedback list without page reload

Story 5: Browse Feedback by Rating As a developer, I want to browse all feedback sorted by highest rating, lowest rating, or most recent, so I can discover the most well-regarded work in the community.

- Sort dropdown with three options: Most Recent (default), Highest Rating, Lowest Rating
- Sort applies immediately and resets to page 1
- Feedback list shows project title when viewing all feedback (not scoped to a single project)

Story 6: Paginated Feedback and Bookmarks As a developer, I want paginated feedback and bookmark lists showing 10 items per page with Previous/Next controls, so the pages load quickly even with many records.

- Previous/Next buttons with disabled state at boundaries
- "Showing X–Y of Z records" summary below controls
- Page resets to 1 when filters or sort order changes

Shared - Authentication

Story 1: Register As a new user, I want to register an account with a username, password, and display name, so I can access the platform.

- Form validates all fields are filled
- Duplicate username shows error message

- Auto-login after successful registration

Story 2: Login As a returning user, I want to log in with my credentials, so I can access my bookmarks and leave feedback under my name.

- Error message on incorrect credentials
- Session persists across page refreshes
- Redirects to project list on successful login

Story 3: Logout As a logged-in user, I want to log out from the dropdown menu, so I can secure my session when I am done.

- Logout clears session and returns to login page
- All authenticated features hidden when logged out

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4. Technical Independence

Preshit Ravindra Pimple's Work

Projects collection (`title`, `description`, `techStack[]`, `category`, `githubLink`, `demoURL`, `authorName`, `createdAt`, `updatedAt`) with full React frontend components (ProjectList, ProjectCard, ProjectFilter, ProjectForm, ProjectDetail, StatsPanel), Express routes (`/api/projects`), and MongoDB integration for project CRUD, filtering by stack and category, search by title/description, pagination, and community stats. **Fully functional without the bookmarks and feedback system.**

Gurudatt Pramod Gaonkar's Work

Bookmarks collection (`projectId`, `userName`, `note`, `createdAt`, `updatedAt`) and **Feedback collection** (`projectId`, `userName`, `rating`, `comment`, `createdAt`, `updatedAt`) with full React frontend components (BookmarkList, BookmarkCard, FeedbackList, FeedbackForm, RatingSort, Pagination), Express routes (`/api/bookmarks`, `/api/feedback`), and MongoDB integration for bookmarking with notes, feedback CRUD, sorting by rating/recency, and pagination. **Fully functional without the project listings system.**

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5. Database Schema

projects Collection

```
{
  _id: ObjectId,
  title: String,
  description: String,
  techStack: Array[String],    // ["React", "Node", "MongoDB"]
  category: String,            // Web | Mobile | AI | DevTools | Game | Data
  githubLink: String,
  demoURL: String,
  authorName: String,
  createdAt: Date,
  updatedAt: Date
}
```

feedback Collection

```
{
  _id: ObjectId,
  projectId: String,           // Reference to projects._id
  userName: String,
  rating: Number,              // 1–5
  comment: String,
  createdAt: Date,
  updatedAt: Date
}
```

bookmarks Collection

```
{
  _id: ObjectId,
  projectId: String,           // Reference to projects._id
  userName: String,
  note: String,                // Optional personal note
  createdAt: Date,
  updatedAt: Date
}
```

users Collection

```
{
  _id: ObjectId,
  username: String,            // Unique
  password: String,
  displayName: String,
  createdAt: Date
}
```

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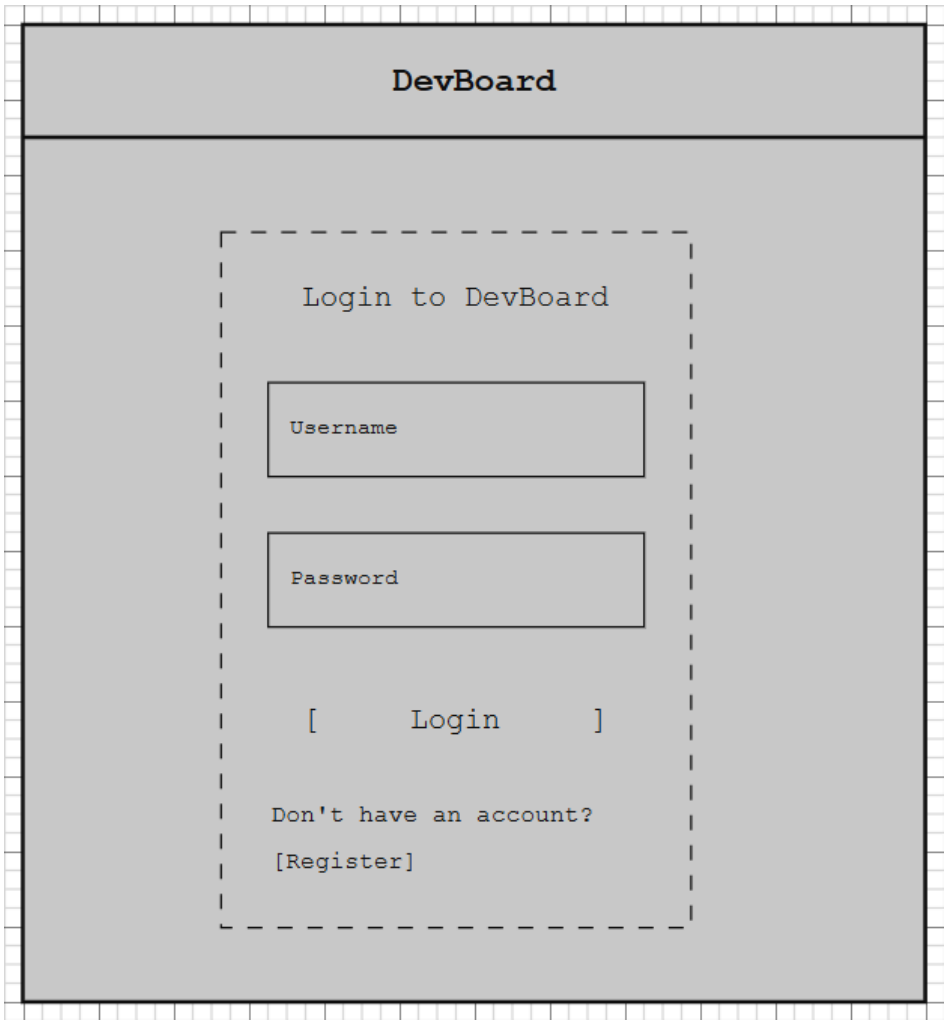
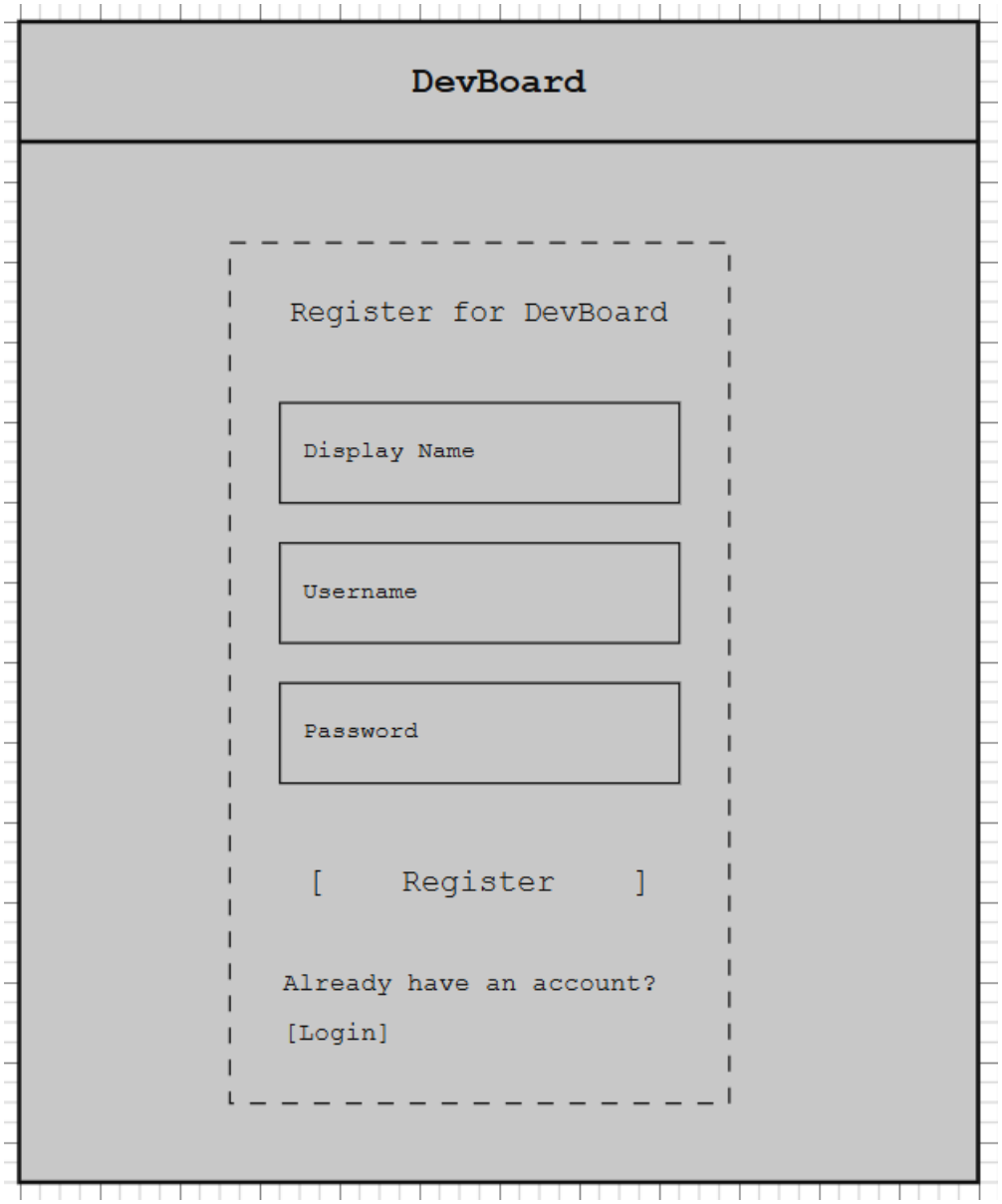
6. Design Mockups

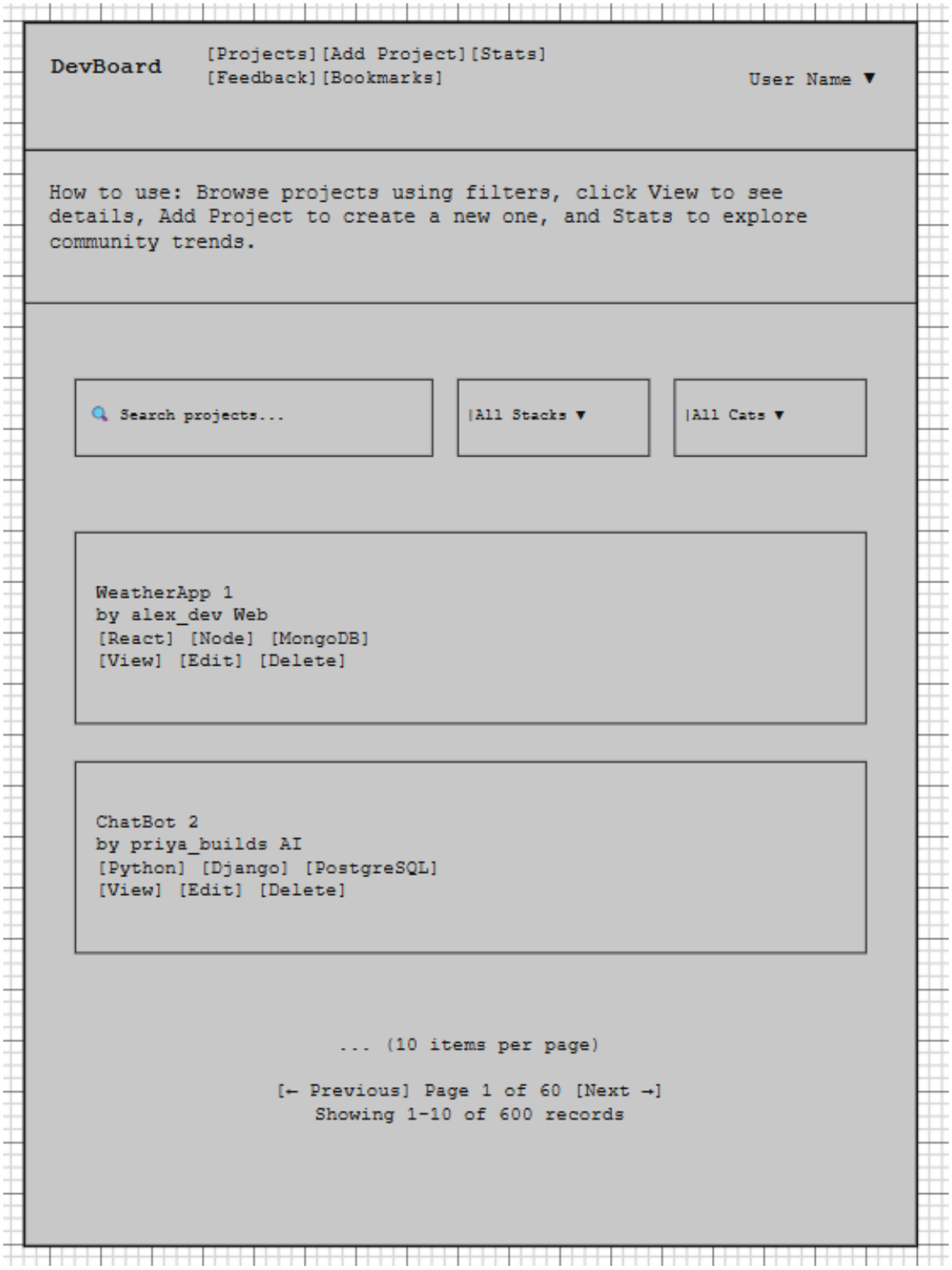
6.1 Login Page



6.2 Register Page



6.3 Project Listing Page (Search, Filter)



6.4 Project Detail Page (with Bookmark, Feedback)

DevBoard [Projects][Add Project][Stats]
[Feedback][Bookmarks] User Name ▼

[← Back]

WeatherApp 1

Author: alex_dev
Category: Web
Description: A full stack web application built with modern technologies.
Tech Stack: [React] [Node] [MongoDB]
GitHub: https://github.com/alex_dev/project-1
Demo: https://project-1.vercel.app
Created: 3/15/2025

[Bookmark This Project] [Add Feedback]

── Feedback (24) ──────────── [Most Recent ▼] ──

jordan_codes ★★★★★
Great project! Clean code.
3/20/2025
[Edit] [Delete]

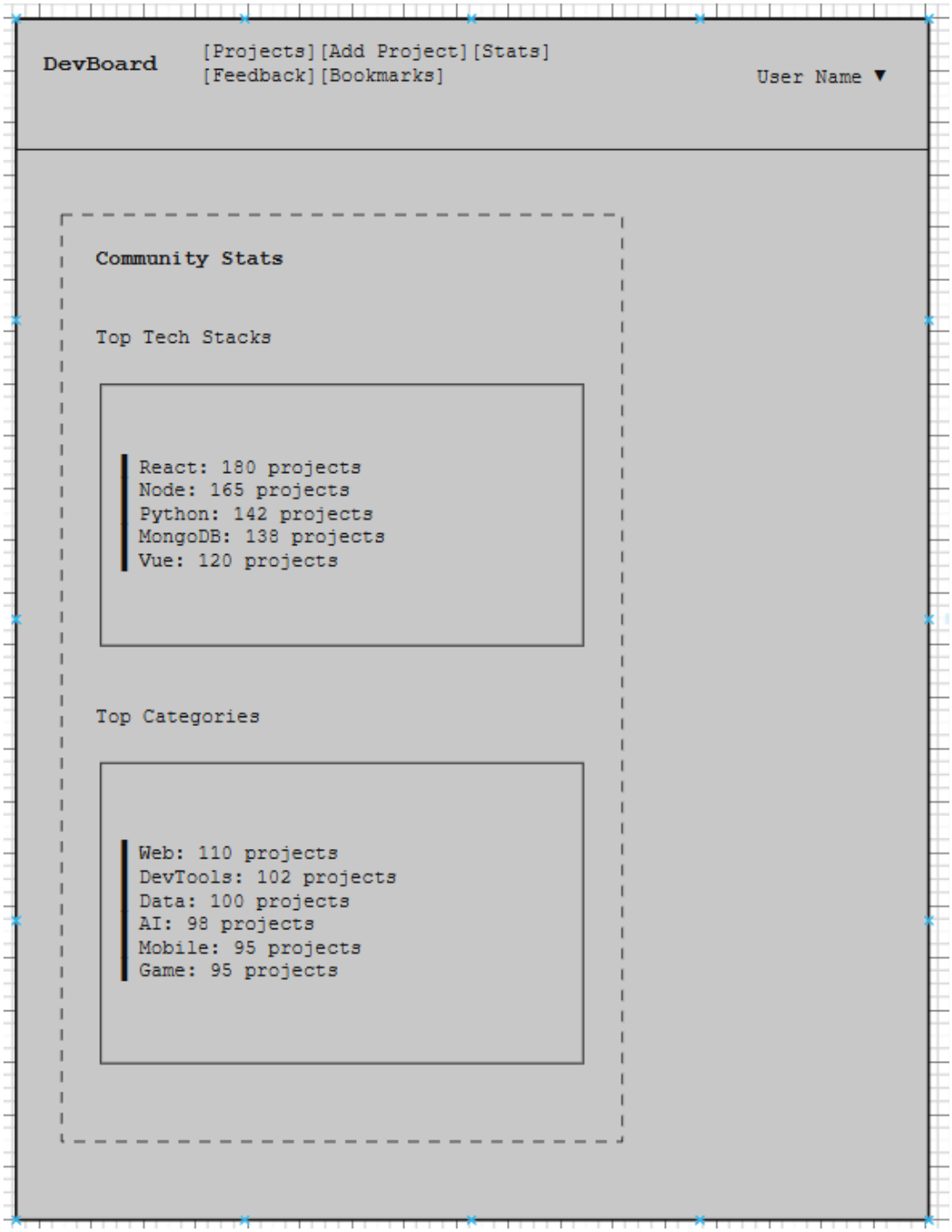
priya_builds ★★★★☆
Love the UI design.
3/18/2025
[Edit] [Delete]

...

[← Previous] Page 1 of 3 [Next →]
Showing 1-10 of 24 records

6.5 Project Create / Edit Form

6.6 Community Stats Panel



6.7 All Feedback Page (with Sort)

```
DevBoard    [Projects][Add Project][Stats]
            [Feedback][Bookmarks]                User Name ▼

All Feedback                                     [Add Feedback]
                                                 [Most Recent ▼]

  alex_dev ★★★★★
  Project: WeatherApp 1
  Great project! Really impressive work.
  3/20/2025
  [Edit] [Delete]

  priya_builds ★★★★☆
  Project: ChatBot 2
  Love the UI design. Very creative idea.
  3/19/2025
  [Edit] [Delete]

                ... (10 items per page)

          [← Previous] Page 1 of 25 [Next →]
             Showing 1-10 of 250 records
```

6.8 Feedback Create / Edit Form

6.9 Bookmarks Page

```
DevBoard    [Projects][Add Project][Stats]
            [Feedback][Bookmarks]                User Name ▼

My Bookmarks (12)

  Project: WeatherApp 1 by alex_dev
  Check this out later
  3/18/2025
  [Edit Note] [Delete]

  Project: ChatBot 2 by priya_builds
  Great reference project
  3/15/2025
  [Edit Note] [Delete]

              ... (10 items per page)

          [← Previous] Page 1 of 2 [Next →]
             Showing 1-10 of 12 records
```

6.10 Bookmark Card — Edit Note Mode

Project: WeatherApp 1 by alex_dev

| Edit your note here...

[Save] [Cancel]

3/18/2025

[Edit Note] [Delete]